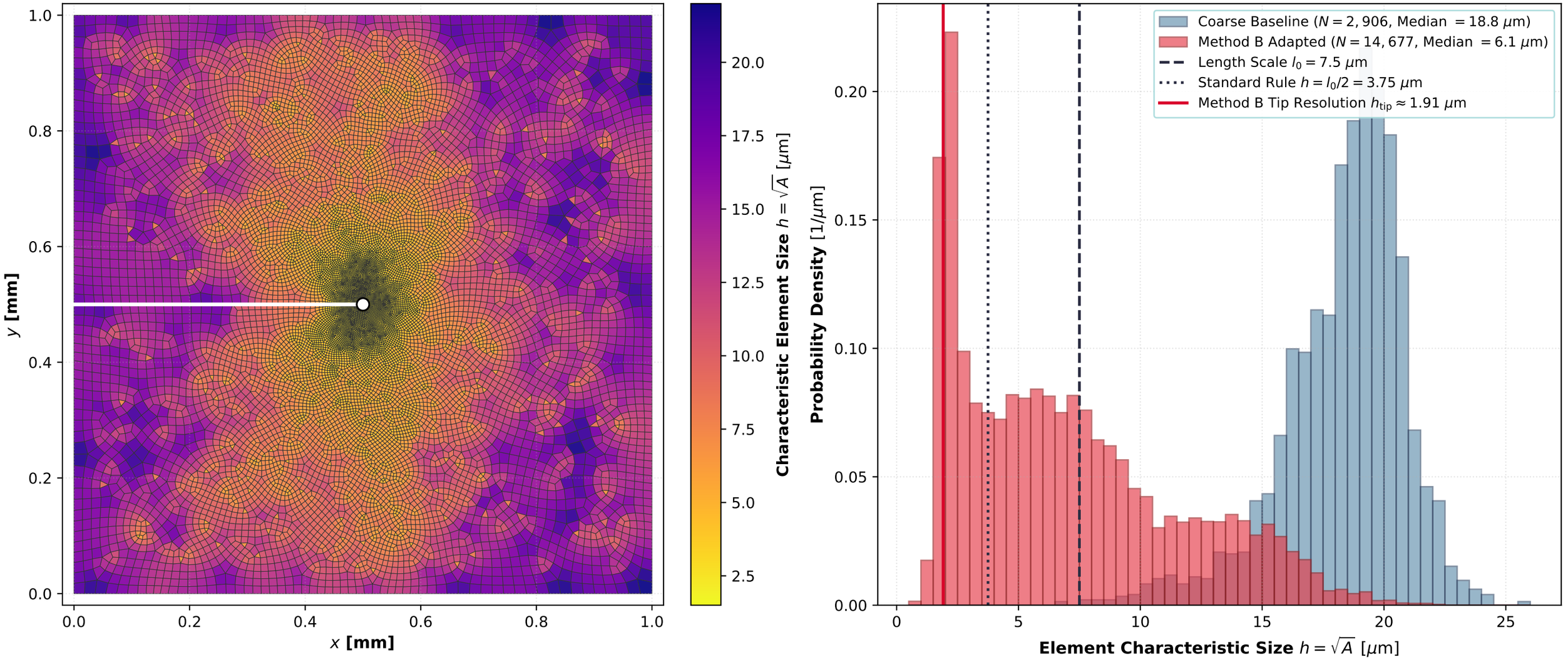

(a) Method B Adapted Mesh: Spatial Element Size Map
Gradient from Tip ($h \approx 1.91$ $\mu$m) to Periphery ($h \approx 22$ $\mu$m)
$y$ [mm]
$x$ [mm]
0.0
0.2
0.4
0.6
0.8
1.0
Characteristic Element Size $h = \sqrt{A}$ [$\mu$m]
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
(b) Discretization Size Distribution & Regularization Thresholds
Comparison of Element Sizing Against Length Scale $l_0 = 7.5$ $\mu$m
Coarse Baseline ($N = 2,906$, Median $= 18.8$ $\mu$m)
Method B Adapted ($N = 14,677$, Median $= 6.1$ $\mu$m)
Length Scale $l_0 = 7.5$ $\mu$m
Standard Rule $h = l_0/2 = 3.75$ $\mu$m
Method B Tip Resolution $h_{tip} \approx 1.91$ $\mu$m
Probability Density [1/$\mu$m]
0.00
0.05
0.10
0.15
0.20
0
5
10
15
20
25
Element Characteristic Size $h = \sqrt{A}$ [$\mu$m]